EARTHSCAPE CLIMATE AGENCY

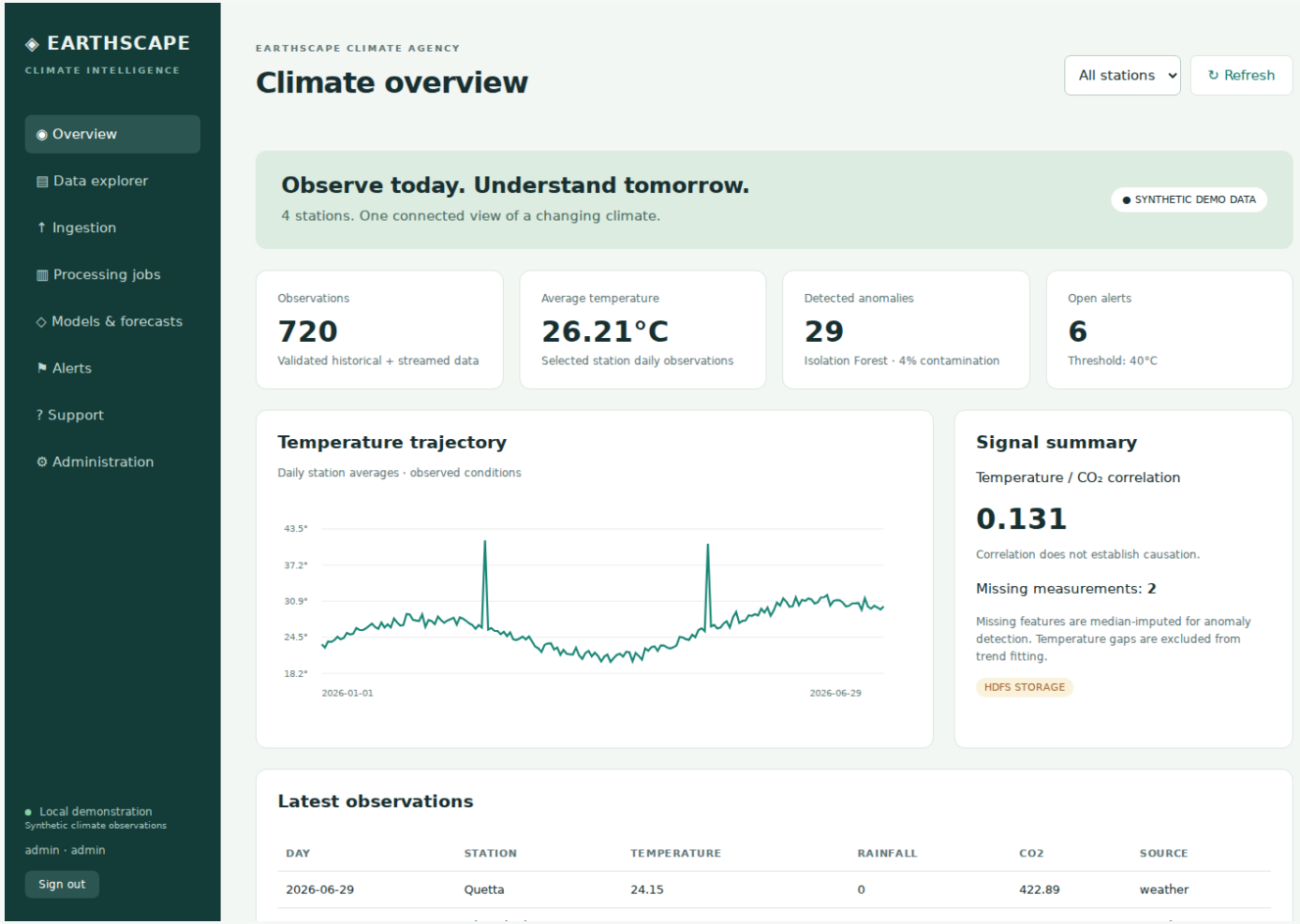
EarthScape

Climate data analysis and monitoring
Project demonstration and technical overview

The project

EarthScape brings climate observations into one application.

Users import records, explore station trends, inspect predictions and respond to temperature alerts.

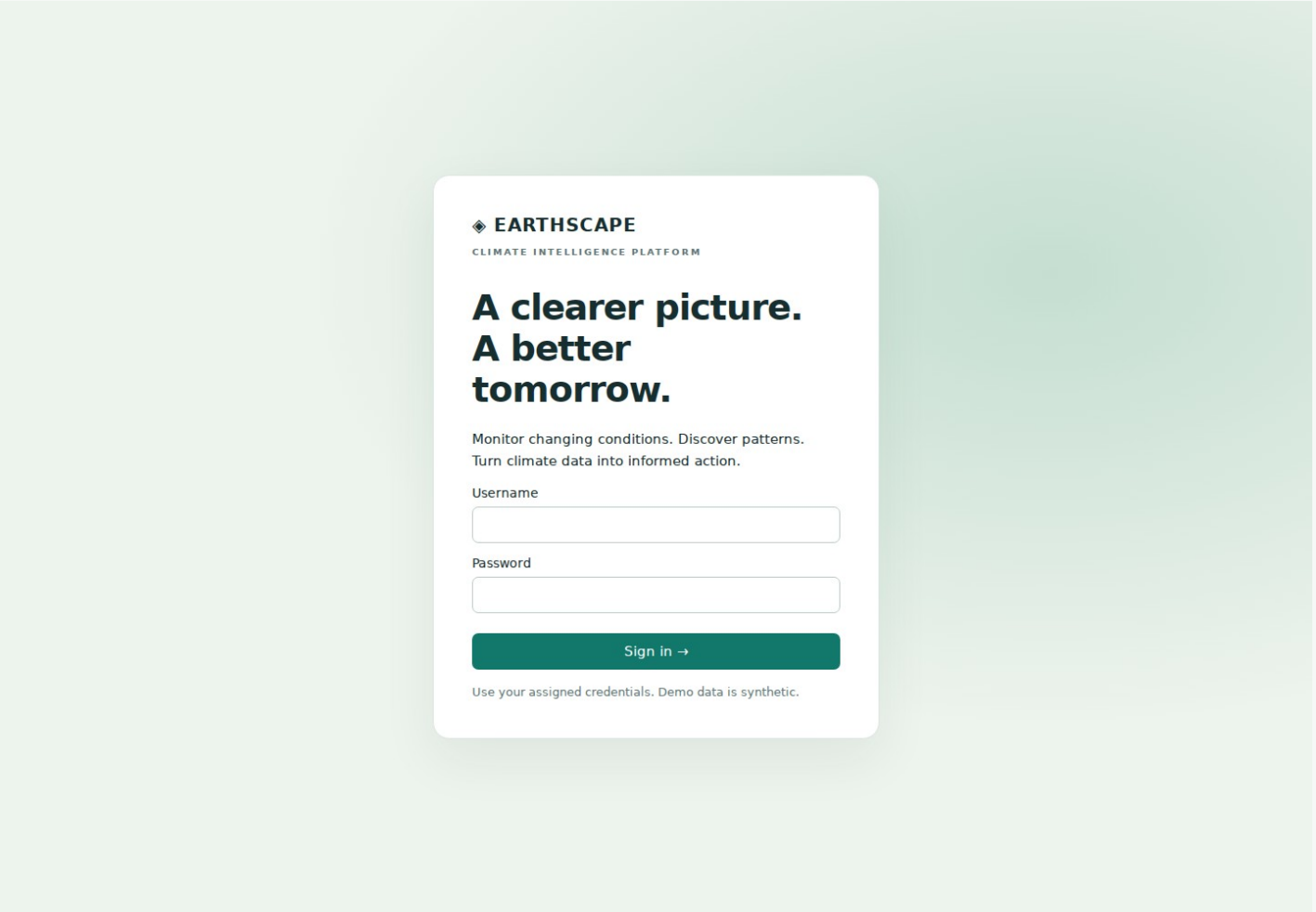


Users and access

Administrator
Accounts, thresholds, backups and audit review

Analyst
Imports, processing and alert acknowledgment

Viewer
Read analysis and submit support tickets

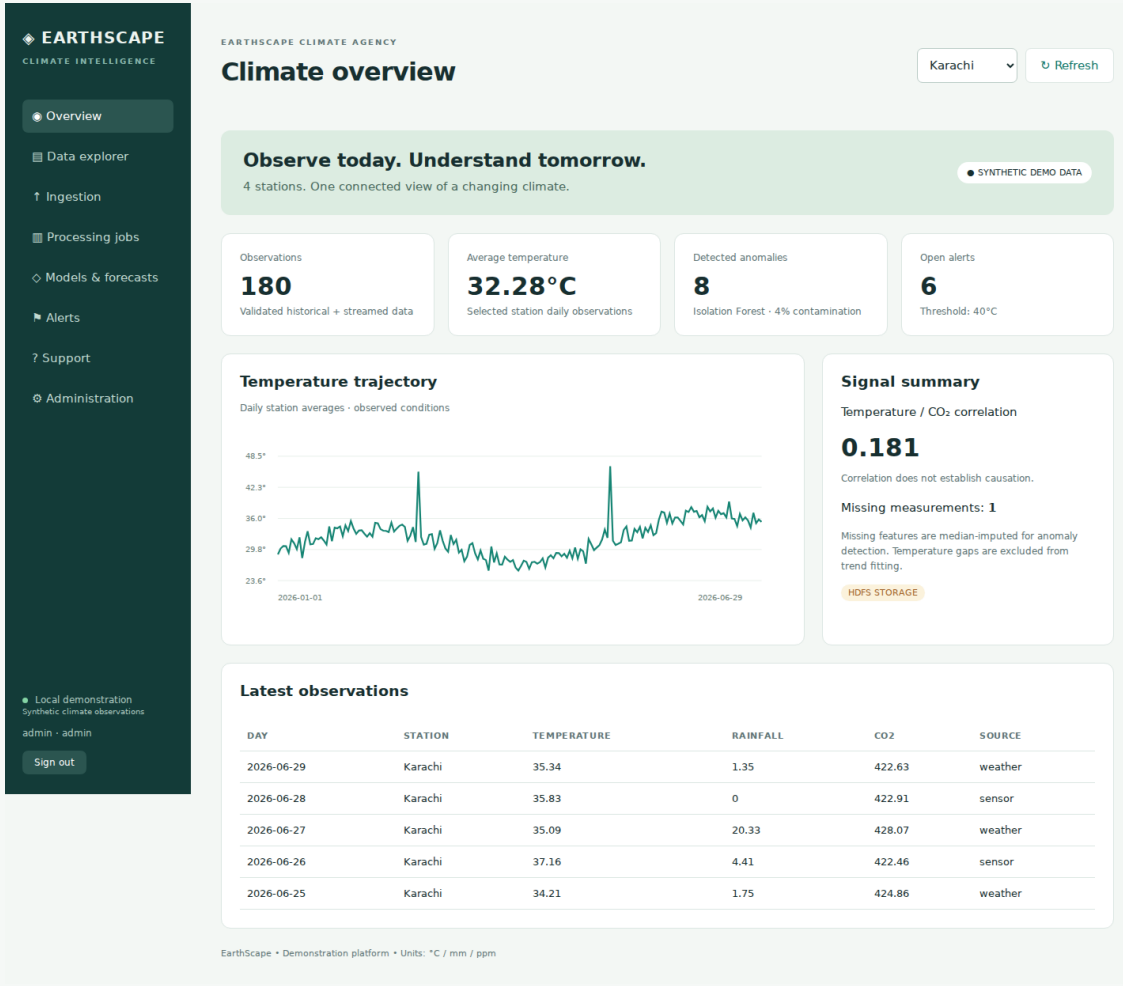


Climate overview

720 synthetic observations
Four stations across 180 dates

Station filters support local comparisons.

The seed data supports repeatable evaluation.

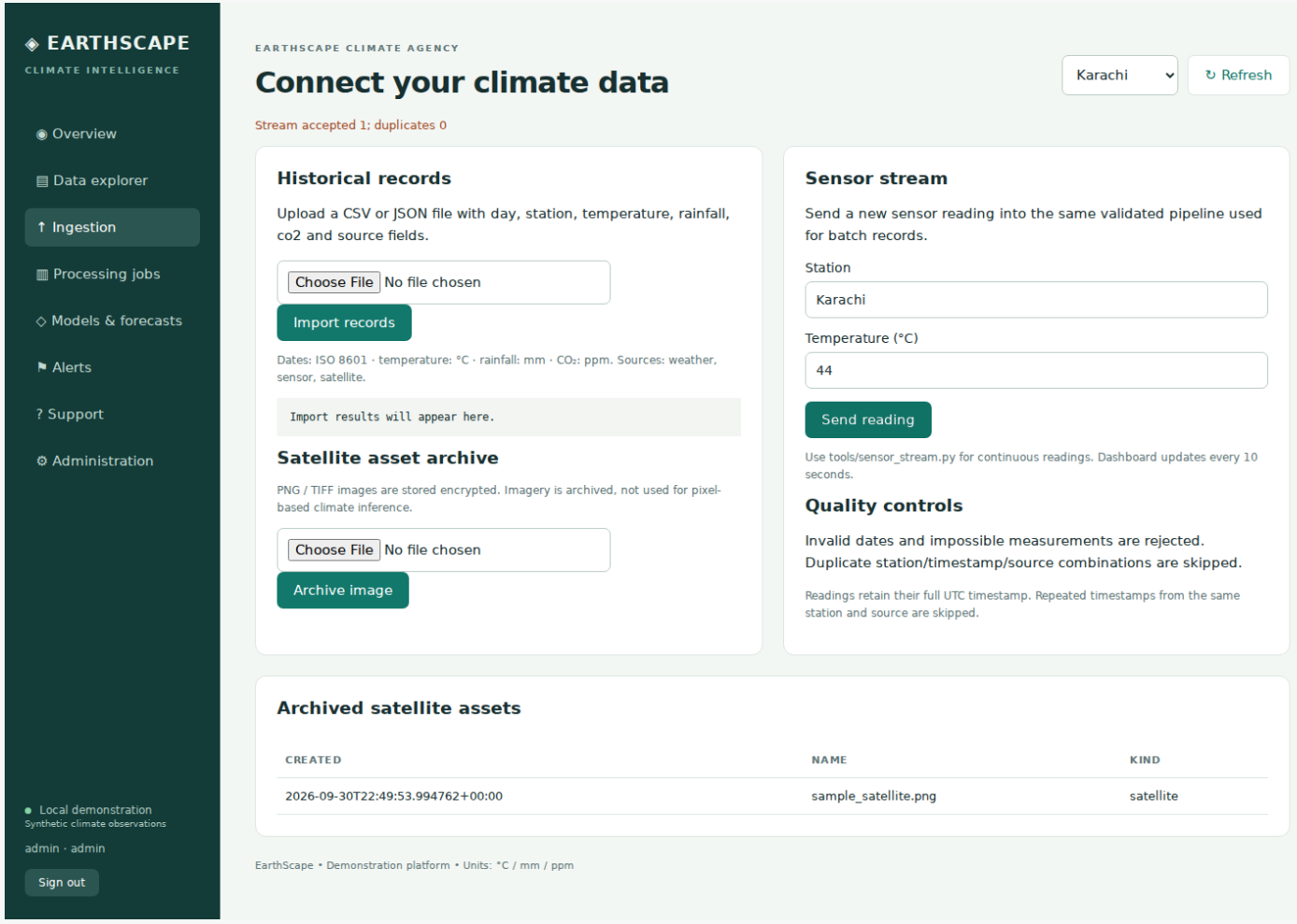


Data ingestion

CSV and JSON records share validation.

Timestamped sensor readings preserve multiple readings per day.

PNG and TIFF satellite assets enter an encrypted archive.

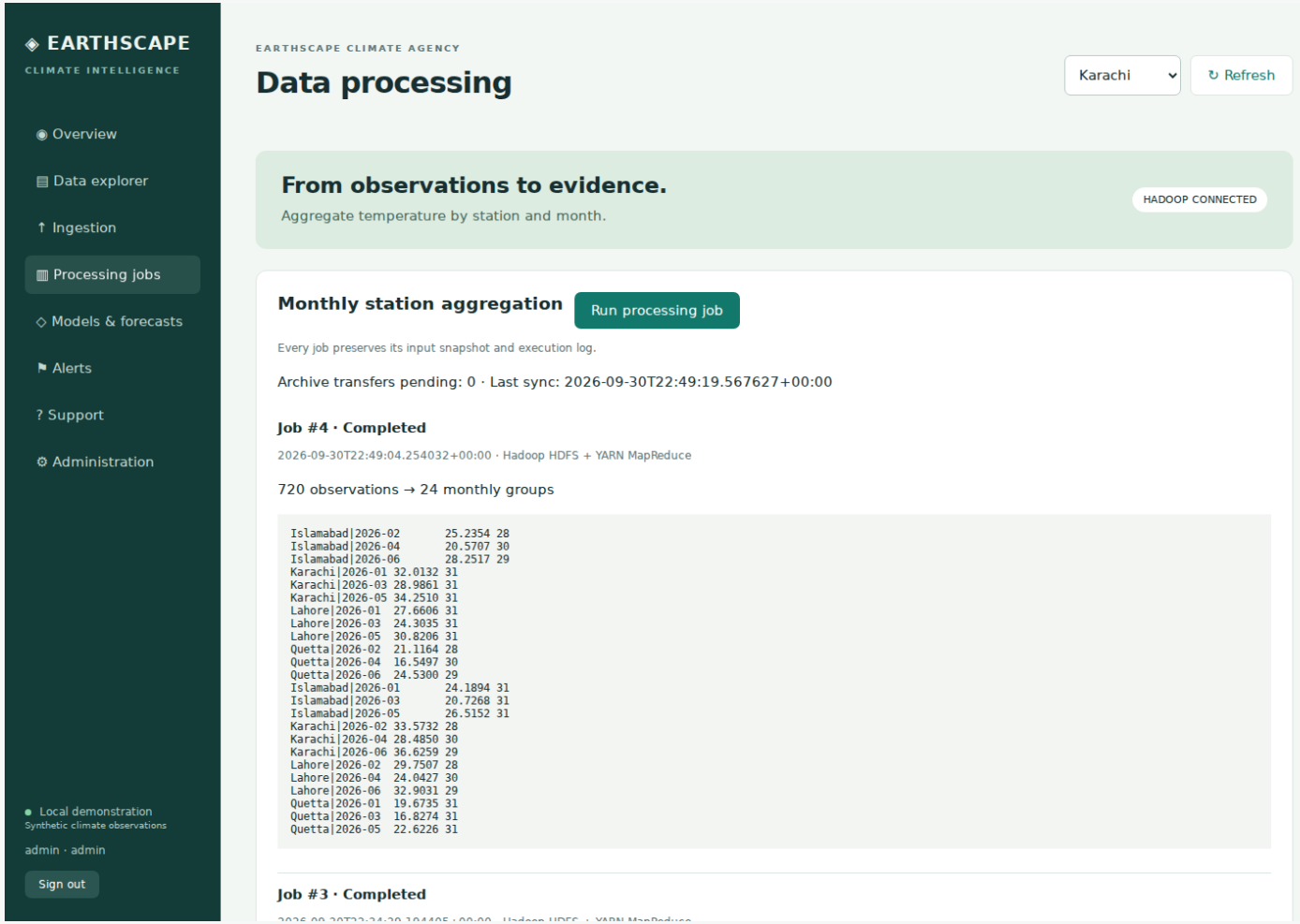


Data processing

Local mode runs mapper and reducer scripts on one computer.

Hadoop mode uses HDFS and YARN with two reducers.

Jobs calculate station/month temperature means.

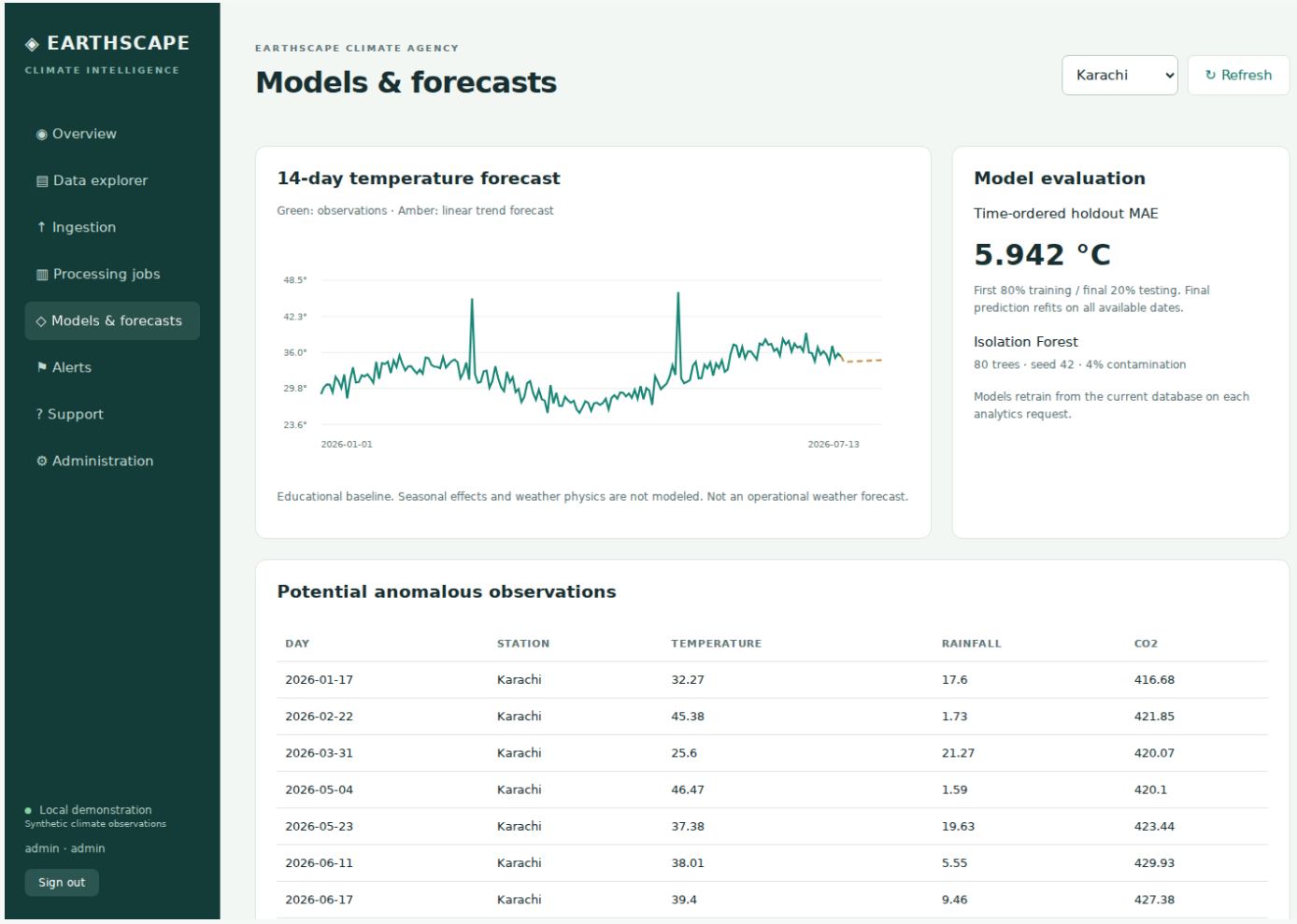


Forecasts and anomalies

A linear regression baseline predicts 14 days.

The final 20% of dates supplies the forecast test set.

Isolation Forest flags unusual observations for review.

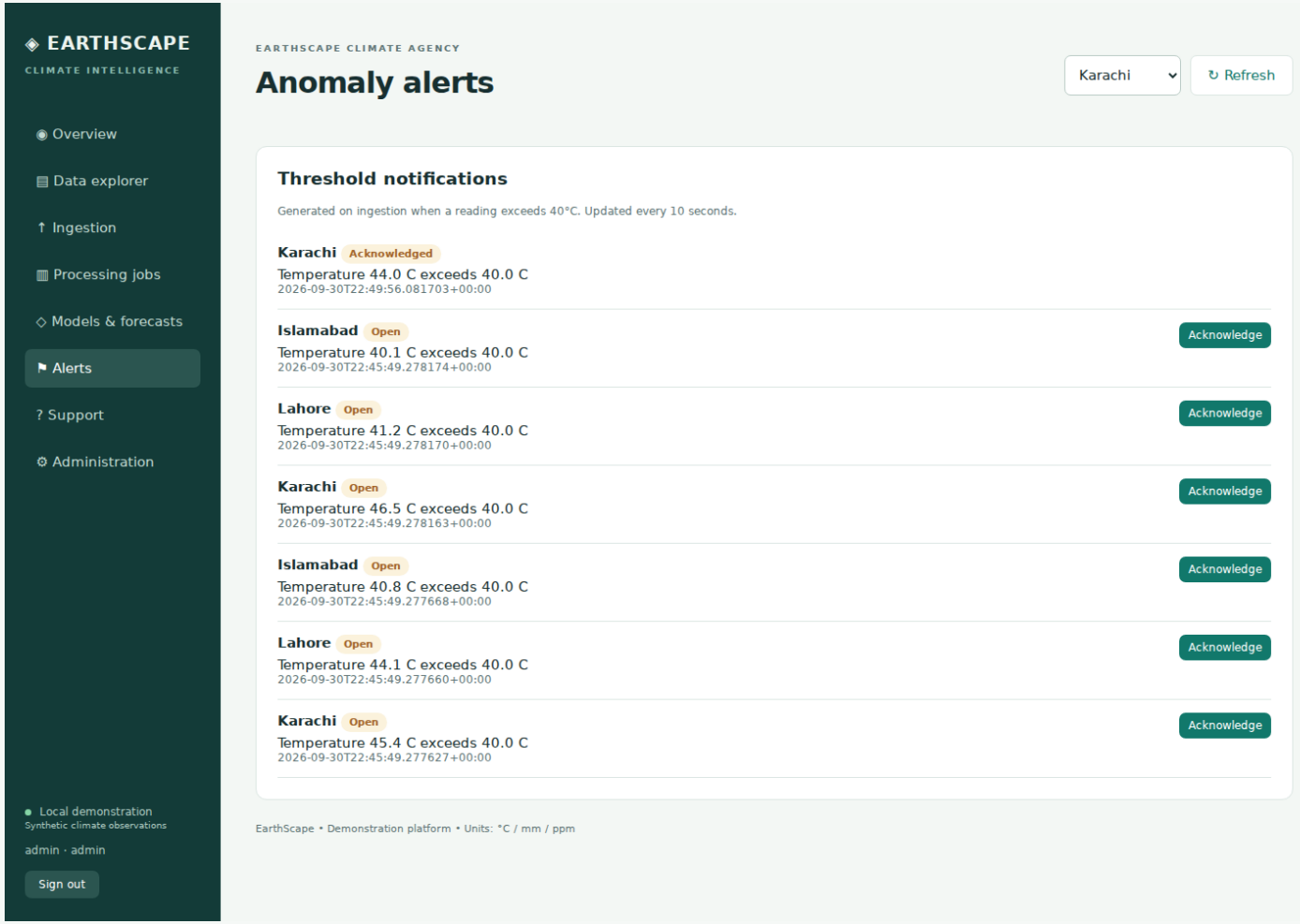


Alerts and support

New readings above the configured threshold create alerts.

Administrators and analysts can acknowledge alerts.

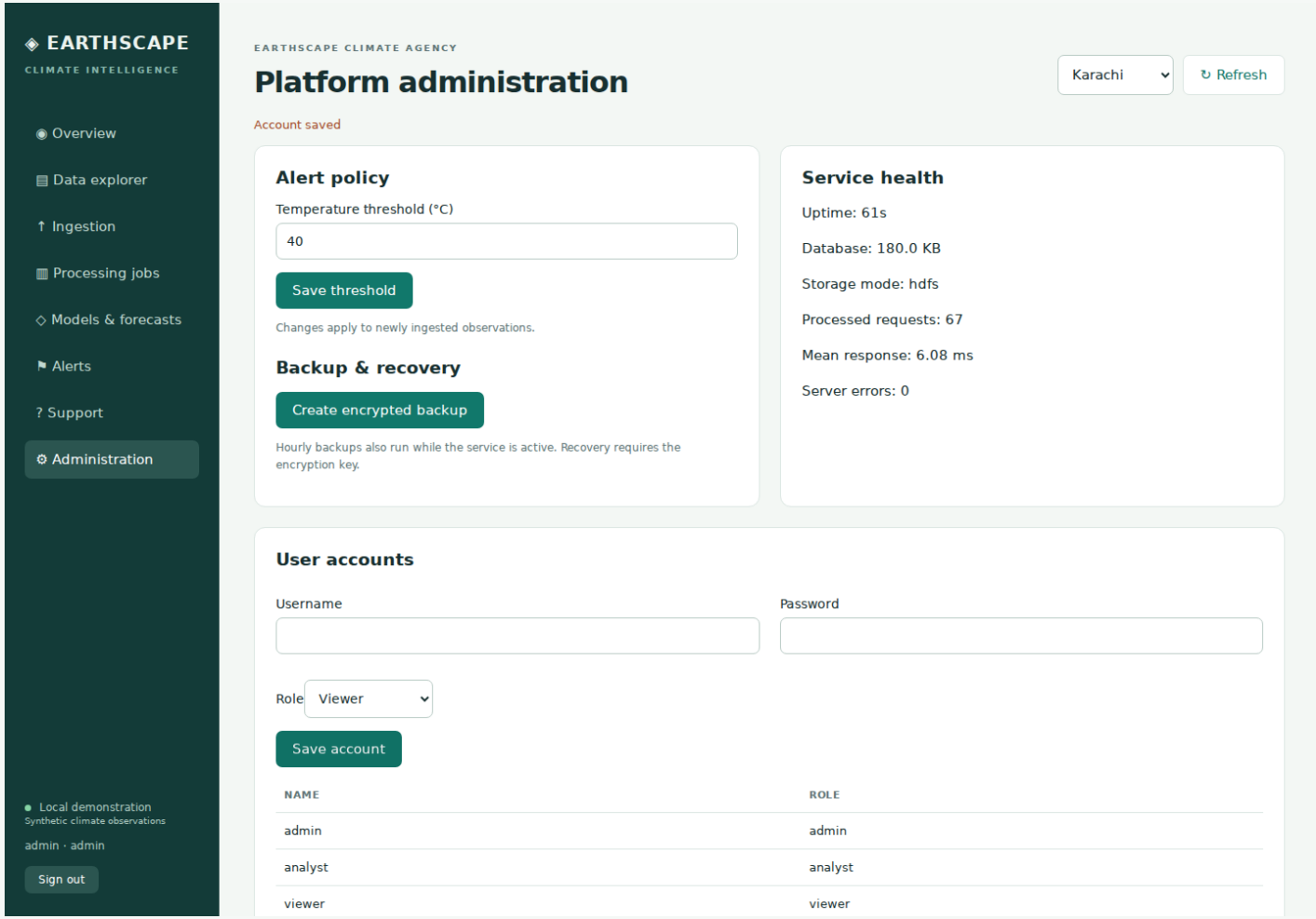
Users submit private tickets for administrator resolution.



Administration

Administrators manage accounts and thresholds.

The interface includes backup actions, process metrics and audit history.

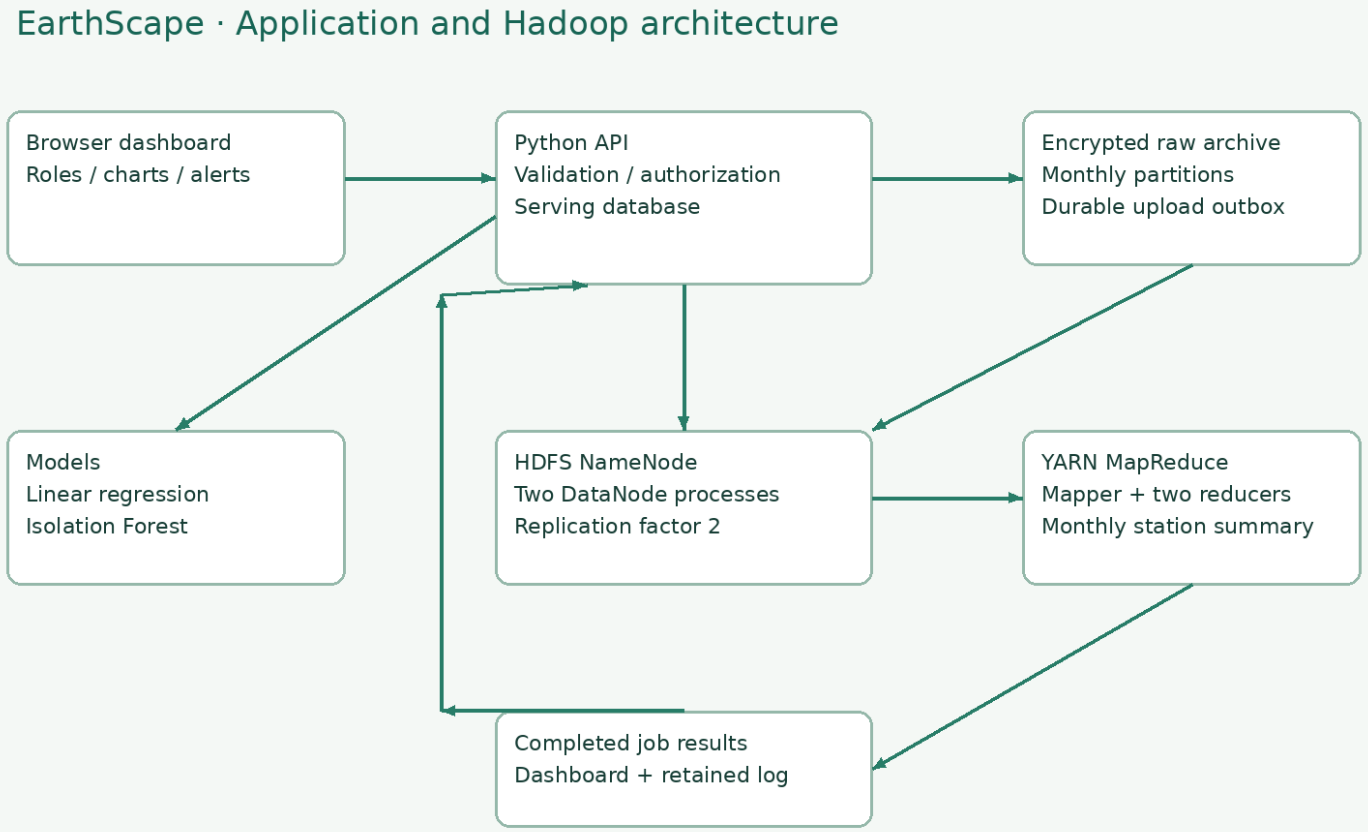


System architecture

The browser uses a Python JSON API.

SQLite serves operational queries.

A background worker manages encrypted archives and Hadoop jobs.

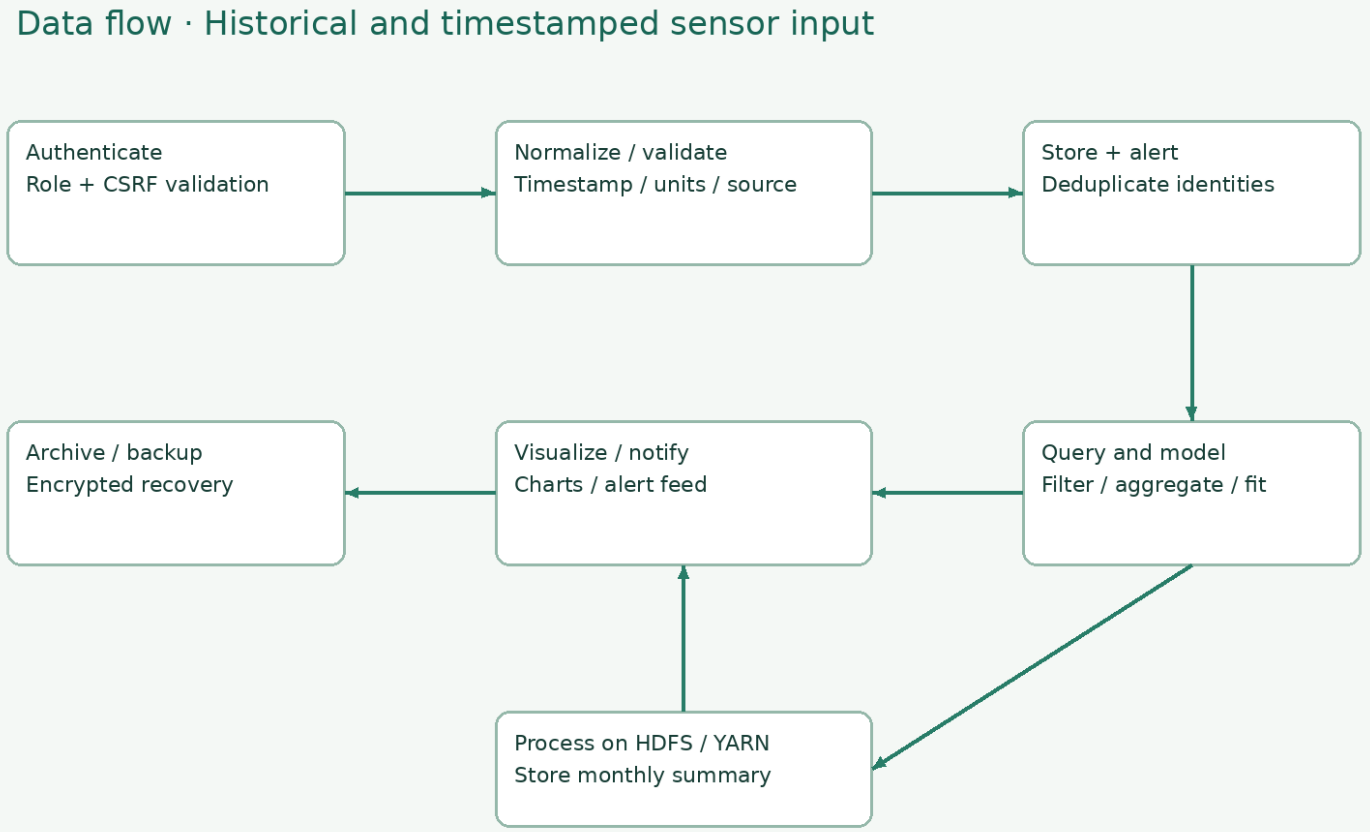


Data flow

Validation normalizes UTC timestamps and checks measurement ranges.

Unique timestamp, station and source identities prevent duplicates.

Archives enter a durable transfer queue.



Security and recovery

Salted PBKDF2 password hashes
Server-side role checks and CSRF tokens
One-hour authenticated sessions
Encrypted archives and backup snapshots

Production requires TLS, managed keys and protected storage.

Setup and demonstration

Windows offline setup
Extract all files, then run START_EARTHSCAPE.bat.
Open http://127.0.0.1:8080.

Full Hadoop setup
Start Docker Desktop, then run START_FULL_STACK.bat.
The first build requires an internet connection.

Validation and limits

18 unit tests passed in this review.
10 HTTP integration checks passed in this review.

The supplied package also retains browser and Hadoop evidence. Windows launch and Docker Desktop were not executed in this Linux review.

Synthetic data and baseline models support an educational demonstration.

Project handover

The package includes the runnable application, Windows runtime, database, source code, reports, diagrams, screenshots and demonstration video.

Add genuine student, batch and group details before the final institutional submission.